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HCMC UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY



HOMEWORK REPORT

COURSE: SOFTWARE TESTING

Assignment: GUI & Usability Testing trên EMS (Event Management System)

STUDENT INFORMATION

Field	Detailed Information
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Task 1

Feature selection

Pool A: FR-02: Login & Account Lock

Test cases:

TC_ID	Title	Type	Priority	Preconditions	Input_Email	Input_Password	Steps	Expected_Result	Notes
TC-LOGIN-01	Valid login - registered user	Positive	High	User account exists and is active	test@eshop.com	Test1234!	1. Navigate to /login 2. Enter email 3. Enter password 4. Click Sign In	Login succeeds; JWT token returned and stored client-side; user redirected to homepage (/)	Baseline happy path
TC-LOGIN-02	Valid login - admin user	Positive	High	Admin account exists	admin@eshop.com	Admin123!	Same steps as TC-LOGIN-01	Login succeeds; JWT token returned; token payload contains role=admin	Confirms role claim for downstream FR-12 access control
TC-LOGIN-03	Incorrect password - 1st failed attempt	Negative	High	Account exists; fail counter = 0	test@eshop.com	WrongPass1!	Submit login once with wrong password	Login rejected; generic error message shown; fail counter increments by exactly 1 (now = 1)	Verifies 'increments by exactly 1 unit' rule
TC-LOGIN-04	Incorrect password - 2nd consecutive failed attempt	Negative	High	Fail counter = 1 (from TC-LOGIN-03)	test@eshop.com	WrongPass2!	Submit login again with wrong password	Login rejected; generic error; fail counter = 2; account NOT yet locked	Boundary case one attempt below lock threshold

TC_ID	Title	Type	Priority	Preconditions	Input_Email	Input_Password	Steps	Expected_Result	Notes
TC-LOGIN-05	Incorrect password - 3rd consecutive failed attempt (triggers lock)	Negative	Critical	Fail counter = 2 (from TC-LOGIN-04)	test@eshop.com	WrongPass3!	Submit login a 3rd consecutive time with wrong password	Account is locked for 30 seconds (demo env); generic lockout error shown; underlying cause is not disclosed	Core FR-02 lockout rule (>= 3 consecutive failures)
TC-LOGIN-06	Correct password submitted while account is locked	Negative	Critical	Account currently locked (immediately after TC-LOGIN-05)	test@eshop.com	Test1234!	Immediately retry login using the CORRECT password while still inside the 30s lock window	Login is still rejected with a lockout message; no JWT is issued	Confirms lock state override; even correct credentials
TC-LOGIN-07	Login succeeds after lockout window expires	Edge	High	Wait 30+ seconds after lock was triggered (post TC-LOGIN-05)	test@eshop.com	Test1234!	Wait 31 seconds, then submit correct credentials	Login succeeds; JWT issued; fail counter resets to 0	Verifies 30s demo expiry and counter reset behavior
TC-LOGIN-08	Non-existent email address	Negative	Medium	Email is not registered in the system	notexist@eshop.com	Test1234!	Submit login form with an unregistered email	Generic error shown (does not reveal 'email not found'); response is indistinguishable from a wrong-password error	Checks account-enumeration / info-leak prevention
TC-LOGIN-09	Empty email field	Negative	Medium	-	(empty)	Test1234!	Leave email field blank, click Sign In	Client-side 'required' validation blocks submission; no network request sent	HTML5 required-field check
TC-LOGIN-10	Empty password field	Negative	Medium	-	test@eshop.com	(empty)	Leave password field blank, click Sign In	Client-side 'required' validation blocks submission; no network request sent	HTML5 required-field check
TC-LOGIN-11	Invalid email format (missing @)	Edge	Medium	-	test.eshop.com	Test1234!	Enter a malformed email, click Sign In	Browser blocks submission via type="email" HTML5 validation	Currently FAIL - field is implemented as type="text" instead of type="email" (known defect)
TC-LOGIN-12	SQL injection attempt in email field	Edge / Security	High	-	' OR '1'='1	anything123	Enter SQLi payload as the email value, submit	Login is rejected as invalid credentials; no authentication bypass occurs; no DB/stack error is exposed to the UI	Validates SEC-05 (parameterized queries)
TC-LOGIN-13	Email case sensitivity	Edge	Low	Account exists as test@eshop.com	TEST@ESHOP.COM	Test1234!	Submit login using the same email in uppercase	Behavior undefined by spec - verify actual system response (commonly expected to succeed, treating	Spec is silent on this; document actual observed behavior as a finding

TC_ID	Title	Type	Priority	Preconditions	Input_Email	Input_Password	Steps	Expected_Result	Notes
TC-LOGIN-14	Password with leading/trailing whitespace	Edge	Low	Account exists	test@eshop.com	Test1234!		email as case-insensitive)	
							Enter the correct password but with extra leading/trailing spaces, submit	Login fails unless the system explicitly trims password input before comparison - verify actual behavior	Confirms no unintended trimming of password field
TC-LOGIN-15	Extremely long string in email field	Edge	Low	-	<500-character string>@eshop.com	Test1234!	Paste a very long string (500+ chars) into the email field, submit	System handles input gracefully (validation error or safe rejection); no client crash or unhandled 500 server error	Boundary / stress test for input length handling

Pool B: FR-09: Coupon

TC_ID	Title	Type	Priority	Preconditions	Input_CouponCode	Input_TotalAmount	Input_LoggedIn	Steps
TC-COUPON-01	Valid percent coupon applied successfully	Positive	High	User logged in; cart total meets threshold	SAVE10	300000	Yes	1. Go to Checkout 2. Enter coupon code 3. Click Áp dụng
TC-COUPON-02	Valid fixed-amount coupon applied successfully	Positive	High	User logged in; cart total meets threshold	BIGBUY	500000	Yes	1. Go to Checkout 2. Enter coupon code 3. Click Áp dụng
TC-COUPON-03	Valid coupon with remaining uses (multi-use code)	Positive	Medium	User logged in; user has used VIP100 0 times (max_uses_per_user = 2)	VIP100	300000	Yes	Enter code and apply
TC-COUPON-04	Non-existent coupon code (C1 fails)	Negative	High	User logged in	NOTREAL99	300000	Yes	Enter a code that does not exist in the DB, click Áp dụng
TC-COUPON-05	Expired coupon rejected (C2 fails)	Negative	High	User logged in; total meets the code's own min_order_amount	EXPIRED	150000	Yes	Enter EXPIRED code (min_order_amount = 100,000 đ, expired_at = 2020-01-01), click Áp dụng
TC-COUPON-06	Order total below coupon's minimum threshold (C3 fails)	Negative	High	User logged in; SAVE10 requires total >= 300,000 đ	SAVE10	200000	Yes	Set cart/edit total to 200,000 đ, enter SAVE10, click Áp dụng
TC-COUPON-07	Guest user attempts to apply a coupon (C4 fails)	Negative	High	User is NOT logged in (no JWT token); total meets threshold	SAVE10	300000	No	As a guest, enter SAVE10 and click Áp dụng
TC-COUPON-08	Coupon already used the maximum number of times (C5 fails)	Negative	High	User logged in; user has already used SAVE10 once (max_uses_per_user = 1)	SAVE10	300000	Yes	Re-apply SAVE10 a 2nd time as the same user

TC_ID	Title	Type	Priority	Preconditions	Input_CouponCode	Input_TotalAmount	Input_LoggedIn	Steps
TC-COUPON-09	Boundary - total exactly equal to min_order_amount	Edge	High	User logged in; BIGBUY min_order_amount = 500,000 đ	BIGBUY	500000	Yes	Set total to exactly 500,000 đ, apply BIGBUY
TC-COUPON-10	Boundary - total one unit below min_order_amount	Edge	High	User logged in; BIGBUY min_order_amount = 500,000 đ	BIGBUY	499999	Yes	Set total to 499,999 đ, apply BIGBUY
TC-COUPON-11	Coupon code entered in lowercase	Edge	Medium	User logged in; total meets threshold	save10	300000	Yes	Type lowercase 'save10', click Áp dụng
TC-COUPON-12	Coupon code with leading/trailing whitespace	Edge	Low	User logged in; total meets threshold	SAVE10	300000	Yes	Type the code with extra spaces around it, click Áp dụng
TC-COUPON-13	Tampered total_amount used to fraudulently satisfy threshold	Edge / Security	Critical	User logged in; REAL cart value is only 100,000 đ (below SAVE10's 300,000 đ threshold)	SAVE10	350000	Yes	1. Manually overwrite the editable 'Tổng tiền thanh toán' field to 350,000 đ (real cart is 100,000 đ) 2. Apply SAVE10 3. Proceed to Xác Nhận Thanh Toán
TC-COUPON-14	Empty coupon code - submit blocked client-side	Edge	Low	User logged in	(empty)	300000	Yes	Leave coupon input blank, observe Áp dụng button
TC-COUPON-15	Multi-use coupon exceeds its own usage cap (C5 fails on 3rd use)	Negative	Medium	User logged in; user has already used VIP100 exactly 2 times (max_uses_per_user = 2)	VIP100	300000	Yes	Attempt to apply VIP100 a 3rd time as the same user

Pool C: FR-17: Coupon CRUD

TC_ID	Title	Type	Priority	Preconditions	Input_Code	Input_Type	Input_DiscountValue	Input_MinOrderAmount	Input_Expired
TC-COUPON-ADM-01	Create valid percent coupon	Positive	High	Logged in as admin; code not yet in use	WELCOME20	percent	20	100000	2099-12-31
TC-COUPON-ADM-02	Create valid fixed-amount coupon	Positive	High	Logged in as admin; code not yet in use	FLASH50K	fixed	50000	200000	2099-06-30

TC_ID	Title	Type	Priority	Preconditions	Input_Code	Input_Type	Input_DiscountValue	Input_MinOrderAmount	Input_Expired
TC-COUPON-ADM-03	Delete an existing coupon	Positive	High	A coupon named WELCOME20 already exists (e.g. from TC-ADM-01)	WELCOME20	-	-	-	-
TC-COUPON-ADM-04	Duplicate coupon code rejected	Negative	High	A coupon with code SAVE10 already exists in the system (sample data)	SAVE10	percent	15	100000	2099-12-31
TC-COUPON-ADM-05	Empty coupon code blocks submission	Negative	Medium	Logged in as admin	(empty)	percent	10	100000	2099-12-31
TC-COUPON-ADM-06	Empty discount value blocks submission	Negative	Medium	Logged in as admin	TESTCODE1	percent	(empty)	100000	2099-12-31
TC-COUPON-ADM-07	Empty expiry date blocks submission	Negative	Medium	Logged in as admin	TESTCODE2	percent	10	100000	(empty)
TC-COUPON-ADM-08	Discount value of zero (not positive)	Edge	High	Logged in as admin	ZERODISC	percent	0	100000	2099-12-31
TC-COUPON-ADM-09	Negative discount value	Edge	High	Logged in as admin	NEGDISC	fixed	-15000	100000	2099-12-31
TC-COUPON-ADM-10	Minimum order amount at boundary zero	Edge	Medium	Logged in as admin	ZEROMIN	percent	10	0	2099-12-31
TC-COUPON-ADM-11	Negative minimum order amount	Edge	High	Logged in as admin	NEGMIN	percent	10	-50000	2099-12-31

TC_ID	Title	Type	Priority	Preconditions	Input_Code	Input_Type	Input_DiscountValue	Input_MinOrderAmount	Input_Expired
TC-COUPON-ADM-12	Max uses per user below minimum (0)	Edge	Medium	Logged in as admin	ZEROUSES	percent	10	100000	2099-12-31
TC-COUPON-ADM-13	Percent discount value exceeding 100%	Edge	Low	Logged in as admin	OVER150	percent	150	100000	2099-12-31
TC-COUPON-ADM-14	Non-admin token blocked from coupon creation API	Negative	Critical	Have a valid JWT for a regular (non-admin) user	APIHACK	percent	10	100000	2099-12-31
TC-COUPON-ADM-15	Coupon code auto-uppercased on input	Edge	Low	Logged in as admin	flashsale	percent	10	100000	2099-12-31

Execution

- Used Playwright default configuration for projects to run on 3 major browsers (Chrome, Firefox, Webkit).
- Added HTML as a reporter to Playwright configuration, along with the `title` option for HTML reporter for each test case.

FR-02

Test ID	Status
TC-01	PASSED
TC-02	PASSED
TC-03	PASSED
TC-04	PASSED
TC-05	PASSED
TC-06	PASSED
TC-07	FAILED
TC-08	PASSED
TC-09	PASSED
TC-10	PASSED
TC-11	PASSED
TC-12	PASSED
TC-13	PASSED
TC-14	PASSED
TC-15	PASSED

FR-09

Test ID	Status
TC-01	FAILED

Test ID	Status
TC-02	FAILED
TC-03	FAILED
TC-04	PASSED
TC-05	PASSED
TC-06	PASSED
TC-07	PASSED
TC-08	FAILED
TC-09	FAILED
TC-10	PASSED
TC-11	FAILED
TC-12	FAILED
TC-13	PASSED
TC-14	PASSED
TC-15	FAILED

The failed test cases were added as bugs to the bug report document.

Test cases 8, 11, 12, and 15 failed to execute correctly because their preconditions and inputs encountered bugs in Bug 02 within the Bug_report.md file.

FR-17

Test ID	Status
TC-01	PASSED
TC-02	PASSED
TC-03	PASSED
TC-04	PASSED
TC-05	PASSED
TC-06	PASSED
TC-07	PASSED
TC-08	FAILED
TC-09	FAILED
TC-10	PASSED
TC-11	FAILED
TC-12	PASSED
TC-13	FAILED
TC-14	PASSED
TC-15	PASSED

The failed test cases were added as bugs to the bug report document.

Assertion pattern used for this feature:

- toHaveURL()
- toBeVisible()/not.toBeVisible()
- toHaveText()
- toHaveAttribute()
- toContainText()
- toHaveCount()
- toHaveValue()
- expect().toBe()/toContain()

AI analysis

FR-01

Overall, the AI performed well in generating high-quality baseline test cases and selecting robust fallback locators where semantic attributes were missing, but its initial implementation suffered from critical structural flaws. Specifically, the AI omitted key edge cases and failed to account for stateful backend constraints, generating a flat list of tests that ignored the failed-login counter and account lockout timer, which led to unintended lockouts and flaky failures. Furthermore, the AI-generated configuration was unsuitable for this feature because it enabled parallel execution (`fullyParallel: true`), causing severe race conditions on the login counter; to resolve these issues, we refined the suite by adding missing scenarios, restructuring stateful lockout cases into sequential flows (`test.describe.serial`), enforcing single-worker execution (`workers: 1`), and integrating automated database resets (`node database.js`) before tests to ensure complete isolation.

FR-09

Similar to FR-02, the AI performed well in the initial setup by generating strong test cases and fallback locators, but its implementation lacked an essential database reset mechanism. Without automatically resetting the database before each test run, executing the test suite sequentially causes coupon usage counters to continuously increment, which prematurely exhausts multi-use caps and inadvertently triggers false-positive failures in downstream test cases that should otherwise pass. To guarantee 100% test reliability and isolate stateful dependencies across coupon validation rules, we enhanced the suite by integrating an automated database re-seeding step (`node database.js`) within the `beforeEach` execution hook.

FR-17

Overall, the AI performed well in establishing the initial test framework similar to previous features, successfully covering primary UI workflows like form submissions and tab navigation; however, its generated script remains flawed as it missed several required test cases from the CSV specification and failed to properly validate currency displays, incorrectly flagging valid VND formatted numbers with comma or dot separators as failures due to rigid string assertions instead of flexible regex handling.

Task 2

[Youtube link](#)

Appendices

Appendix A

[AI Audit Report](#)

[AI Critique](#)

Appendix B

[Self-assessment & Test summary report](#)